AI Exploration Challenge - Stage 2

Medsaver

AI - Generic Medicine Finder

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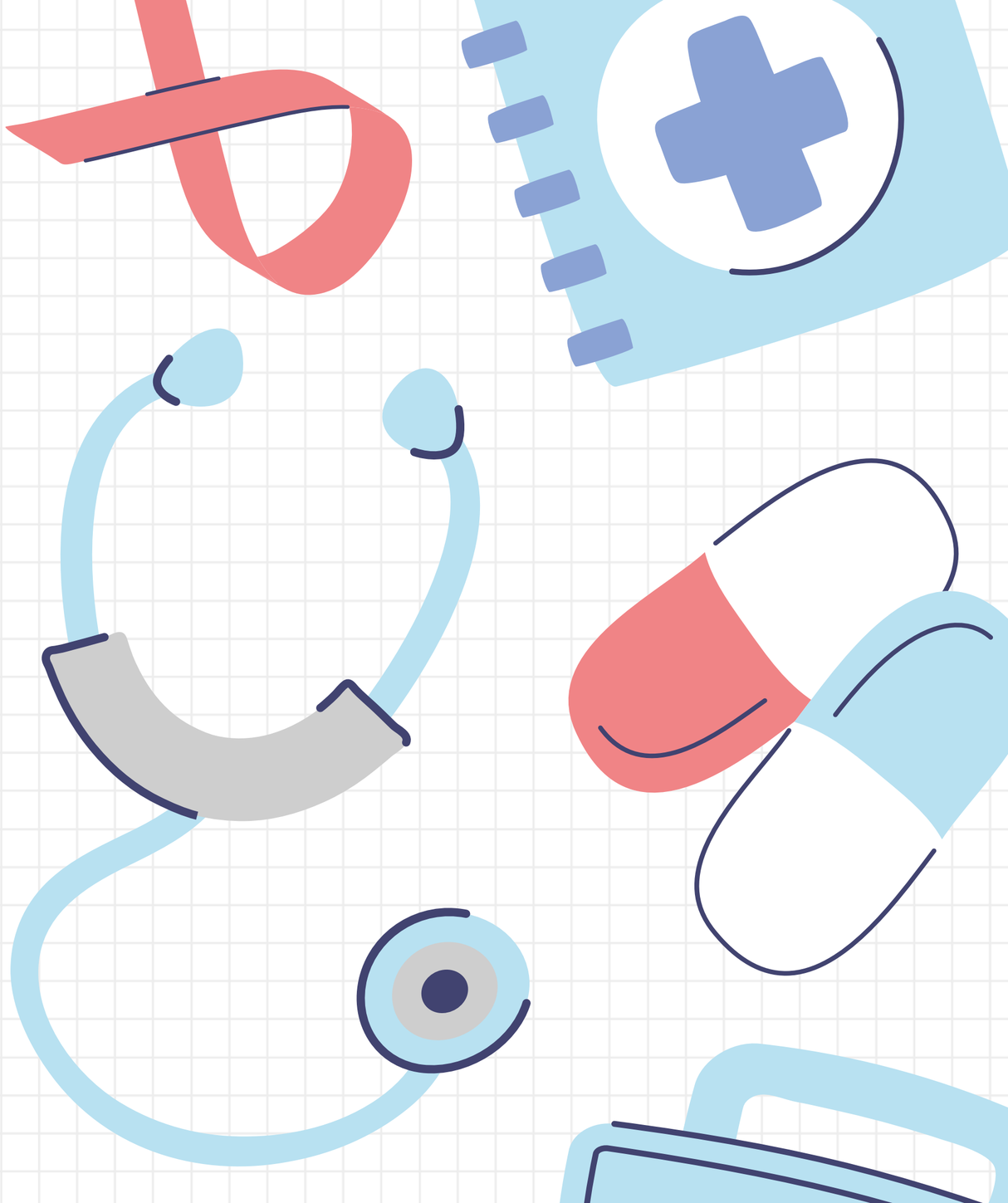
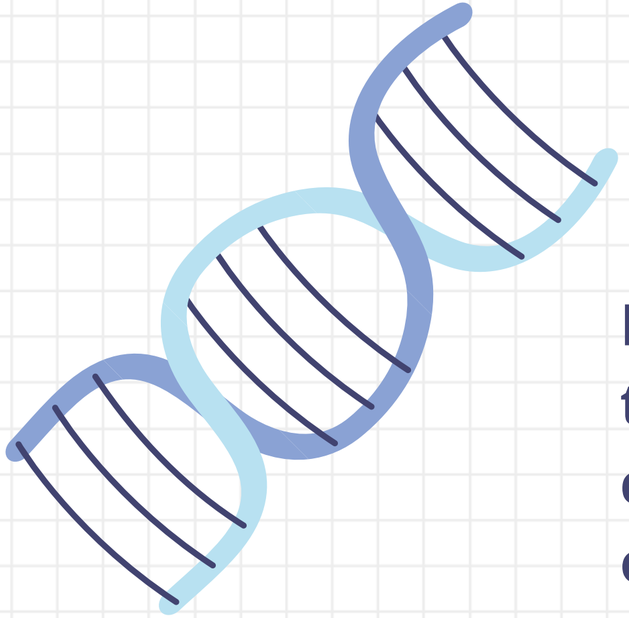


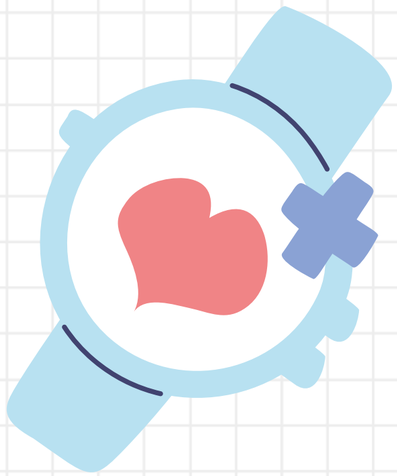


The Problem

Branded medicines in India are typically 5–10x more expensive than their generic equivalents also though branded but cheaper alternative exists.Patients often overpay due to a lack of accessible information.

“Traditional rule-based or keyword search systems struggle because they rely on exact matches, while prescriptions often contain variations, abbreviations, and inconsistent formatting. As a result, patients depend heavily on pharmacists for alternatives, where decisions may vary based on experience, availability, and pricing—leading to limited awareness of more affordable generic options.”

Solution

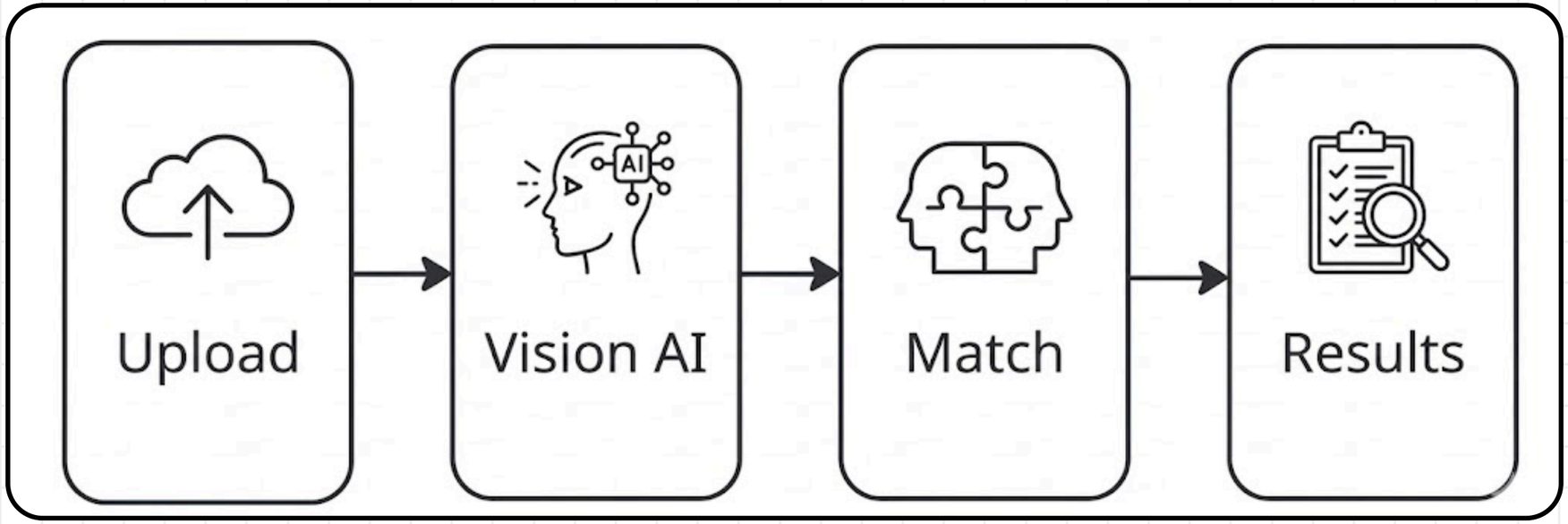


Medsaver is an AI-powered generic medicine finder that helps patients in India save money on medications. Users simply upload a photo of their prescription, and the system uses a Vision LLM to read even illegible handwritten prescriptions, extract the medicine names and their active salt compositions, and match them against a database of 2,052+ government-backed Jan Aushadhi generic drugs and a dataset cotaining 2.5 lakhs+ medicines — returning the cheaper branded alternatives, generic name, pack size, and price in seconds.

Medicine names can appear in many branded forms, abbreviations, or slightly incorrect spellings (especially after OCR). A Generative AI model can understand context, normalize drug names, and map brand medicines to their correct generic salts even when the text is imperfect.

How it works

- **Upload: Patient uploads prescription photo**
- **Vision AI: Llama 4 Scout reads and extracts medicine names and salts.**
- **Match: Cross-reference 2052+ Jan Aushadhi drugs with fuzzy matching.**
- **Results: Returns generic name, pack, size, MRP & per-unit price.**



Datasets

PMBJP

This dataset is a government-maintained catalog of affordable generic medicines, containing details like composition (salt), dosage, form, and price. It enables mapping branded medicines to their equivalent low-cost generics for direct price comparison. This makes it a reliable foundation for identifying safe, cost-effective alternatives.

A-Z Medicines India

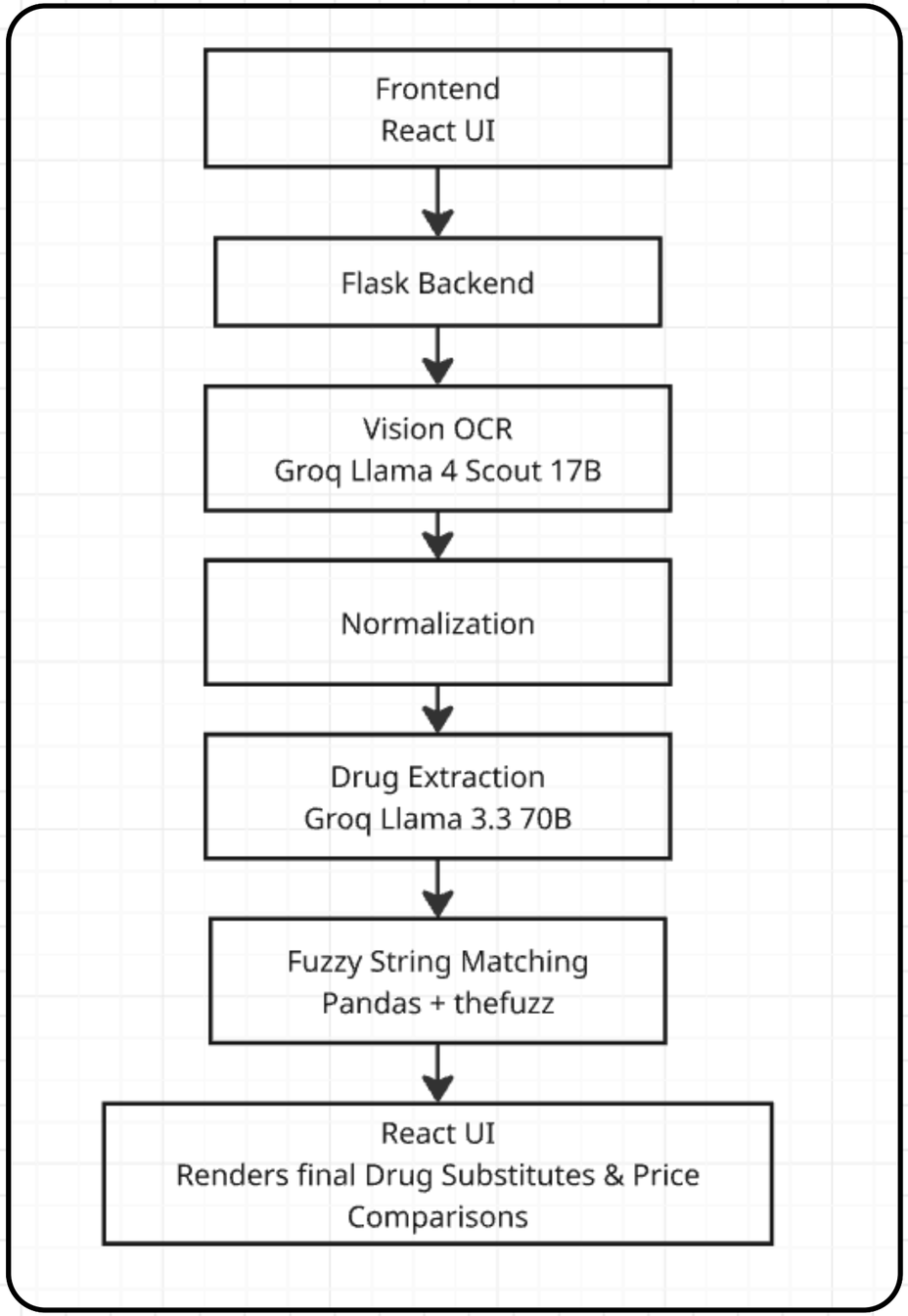
The file has data which contains all possible medicines that we were able to find out during our research on finding out medicine's details such as compositions of medicine, type of medicines, there market availability, pricing and many other things.

Tech Pipeline

pandas

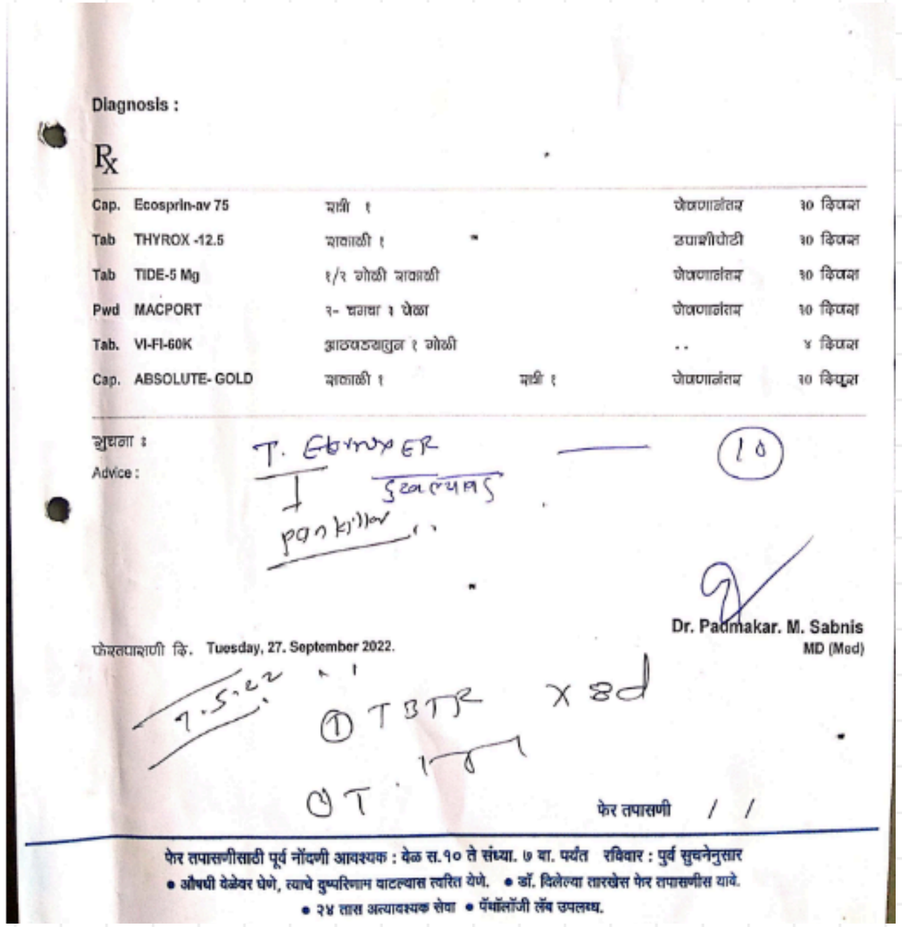
Flask

groq™



Medsaver

Input



Diagnosis :

Rx

Cap.	Ecosprin-av 75	माती १		जेवणानंतर	१० दिवस
Tab	THYROX -12.5	सकाळी १		उपाशीपोटी	१० दिवस
Tab	TIDE-5 Mg	१/२ गोळी सकाळी		जेवणानंतर	१० दिवस
Pwd	MACPORT	२- चमचा १ पेला		जेवणानंतर	१० दिवस
Tab.	VI-FI-60K	आठवड्यातून १ गोळी		..	४ दिवस
Cap.	ABSOLUTE- GOLD	सकाळी १	माती १	जेवणानंतर	१० दिवस

सूचना :
Advice :

T. Ebrinxer
I
pankillar

Dr. Padmakar. M. Sabnis
MD (Med)

फेरतपासणी कि. Tuesday, 27. September 2022.

7.5.22

① TBTR x 8d

फेर तपासणी / /

फेर तपासणीसाठी पूर्व नोंदणी आवश्यक : वेळ स.१० ते संध्या. ७ वा. पर्यंत रविवार : पुर्व सुचनेनुसार
• औषधी घेऊनच घेणे, त्याचे दुष्परिणाम वाटल्यास त्वरित येणे. • डॉ. दिलेल्या तारखेस फेर तपासणीस यावे.
• २४ तास अत्यावश्यक सेवा • पॅथॉलॉजी लॅब उपलब्ध.

Output

THYROX-12.6 Exact match

Levothyroxine 12.6mcg

Thyroid hormone | Tablet

Prescribed Brand

Name	Manufacturer	Pack size	MRP	Per unit
Thyrox 12.5 Tablet	Macleods Pharmaceuticals Pvt Ltd	bottle of 100 tablets	₹167.12	₹1.67

Cheaper Branded Alternatives

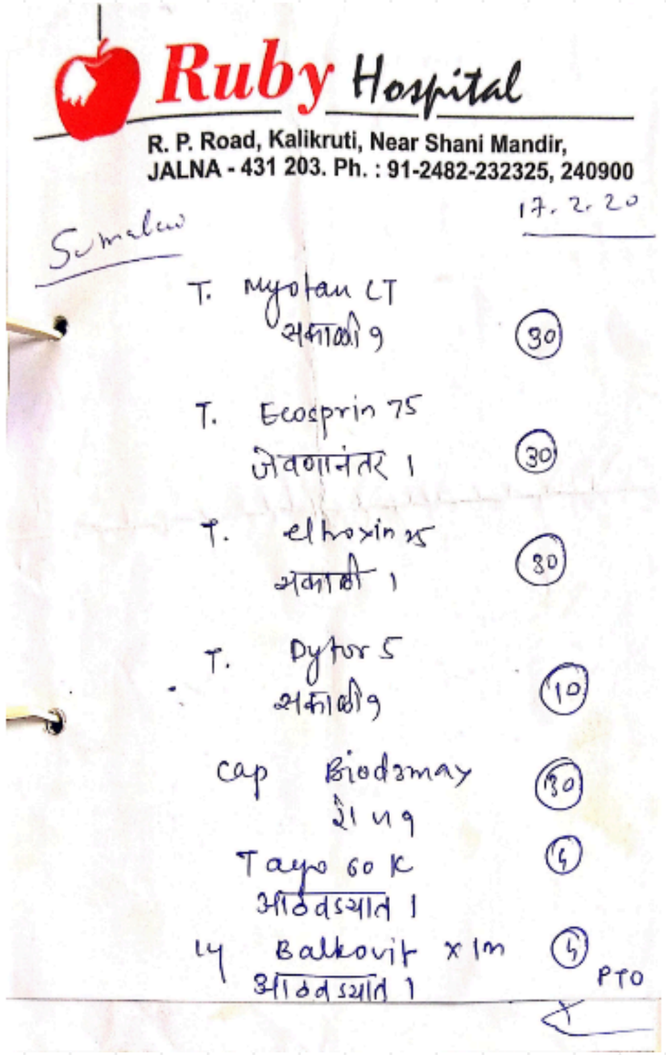
Name	Manufacturer	Pack size	MRP	Per unit
Cheapest Thyroxium 12.5mcg Tablet	Elder Projects Ltd	bottle of 100 tablets	₹63.50	₹0.64
Thyrocut 12.5 Tablet	Care Formulation Labs Pvt Ltd	strip of 100 tablets	₹69.75	₹0.70
Thyropax 12.5mcg Tablet	Alteus Biogenics Pvt Ltd	bottle of 100 tablets	₹75.04	₹0.75
Thyrotas 12.5 Tablet	Intas Pharmaceuticals Ltd	bottle of 50 tablets	₹53.63	₹1.07
Thyroquik 12.5mcg Tablet	Vats Biotech Pvt Ltd	bottle of 100 tablets	₹107.00	₹1.07

Jan Aushadhi Generics

Generic name	Pack size	MRP	Per unit
Best match Levo-Thyroxine Sodium Tablets IP 100 mcg	100's	₹60.00	₹0.60

💡 Switch to Jan Aushadhi — save **₹1.07/unit** (64.1% cheaper than prescribed brand)

Medsaver



Ruby Hospital
R. P. Road, Kalikruti, Near Shani Mandir,
JALNA - 431 203. Ph. : 91-2482-232325, 240900

17. 2. 20

Sumalw

T. Myotan CT
सकाळी 9 — 30

T. Ecosprin 75
जेवणानंतर 1 — 30

T. eltroxin 25
सकाळी 1 — 30

T. Dytor 5
सकाळी 9 — 10

Cap Biodzmay
रो 14 — 30

Tayo 60 K
आठवड्यात 1 — 4

14 Balkovit x 1m
आठवड्यात 1 — 4

PTO

Ecosprin 75 Exact match
Aspirin 75mg
Antiplatelet | Tablet

Prescribed Brand

Name	Manufacturer	Pack size	MRP	Per unit
Ecosprin 75 Tablet	USV Ltd	strip of 14 tablets	₹5.33	₹0.38

Cheaper Branded Alternatives

Name	Manufacturer	Pack size	MRP	Per unit
Cheapest Sprin 75mg Tablet	Alkem Laboratories Ltd	strip of 10 tablets	₹2.05	₹0.20
CV Sprin 75mg Tablet	Cadila Pharmaceuticals Ltd	strip of 10 tablets	₹2.21	₹0.22
Aspidot 75mg Tablet	Three Dots Lifescience	strip of 10 tablets	₹2.43	₹0.24
ASA 75mg Tablet	Zydus Cadila	strip of 14 tablets	₹3.70	₹0.26
E Prin 75mg Tablet	Torrent Pharmaceuticals Ltd	strip of 14 tablets	₹3.72	₹0.27

Jan Aushadhi Generics

Generic name	Pack size	MRP	Per unit
Best match Aspirin Enteric Coated Tablets IP 75mg	14's	₹2.82	₹0.20
Clopidogrel 75mg, Aspirin 75mg Tablets IP	10's	₹19.69	₹1.97

💡 Switch to Jan Aushadhi — save **₹0.18/unit** (47.4% cheaper than prescribed brand)

Tayo 60 K Exact match
Vitamin B12 60mcg
Vitamin supplement | Tablet

Prescribed Brand

Brand not found in A-Z database

Cheaper Branded Alternatives

Name	Manufacturer	Pack size	MRP	Per unit
Cheapest Bvigor Sublingual tablet Cherry	Himerus Medical Solution Llp	strip of 10 Sublingual tablet	₹98.00	₹9.80

Jan Aushadhi Generics

Generic name	Pack size	MRP	Per unit
Best match Thiamine 100mg, Pyridoxine Hydrochloride 100mg, Cyanocobalamin 1000mcg per 3 ml Injection	3 ml	₹7.50	₹2.50
Folic Acid 15mg, Cyanocobalamine 500mcg, Nicotinamide 200mg Injection	10 ML	₹37.50	₹3.75

Platform	Workflow
Netmeds	No automatic generic suggestions,manual entering
Truemeds	Separate seach for each medicine to find substituents no upload facility
Pharmeasy	No transparency, no direct comparison
Tata1mg	No transparency, no direct comparison

Future Scope

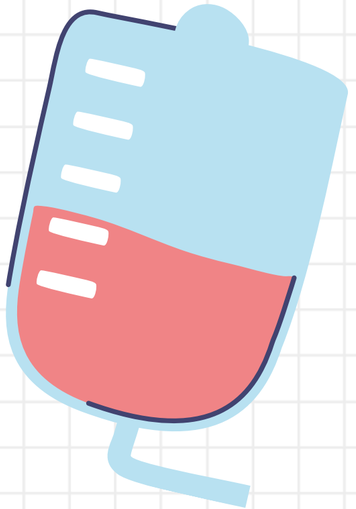
Wider Coverage

- Expand database from 2052 to 100,000+ using multiple datasets.
- Add PDF parsing support.
- Integrate live pharmacy APIs for real-time price checking.

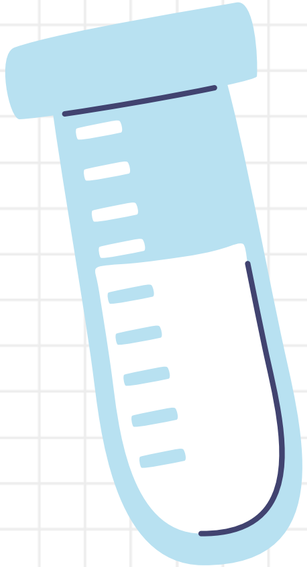
Smart Features

- Launch a React Native mobile app for on-the-go access.
- Introduce RAG based symptom search.
- Add multilingual support for region Indian languages.

Conclusions



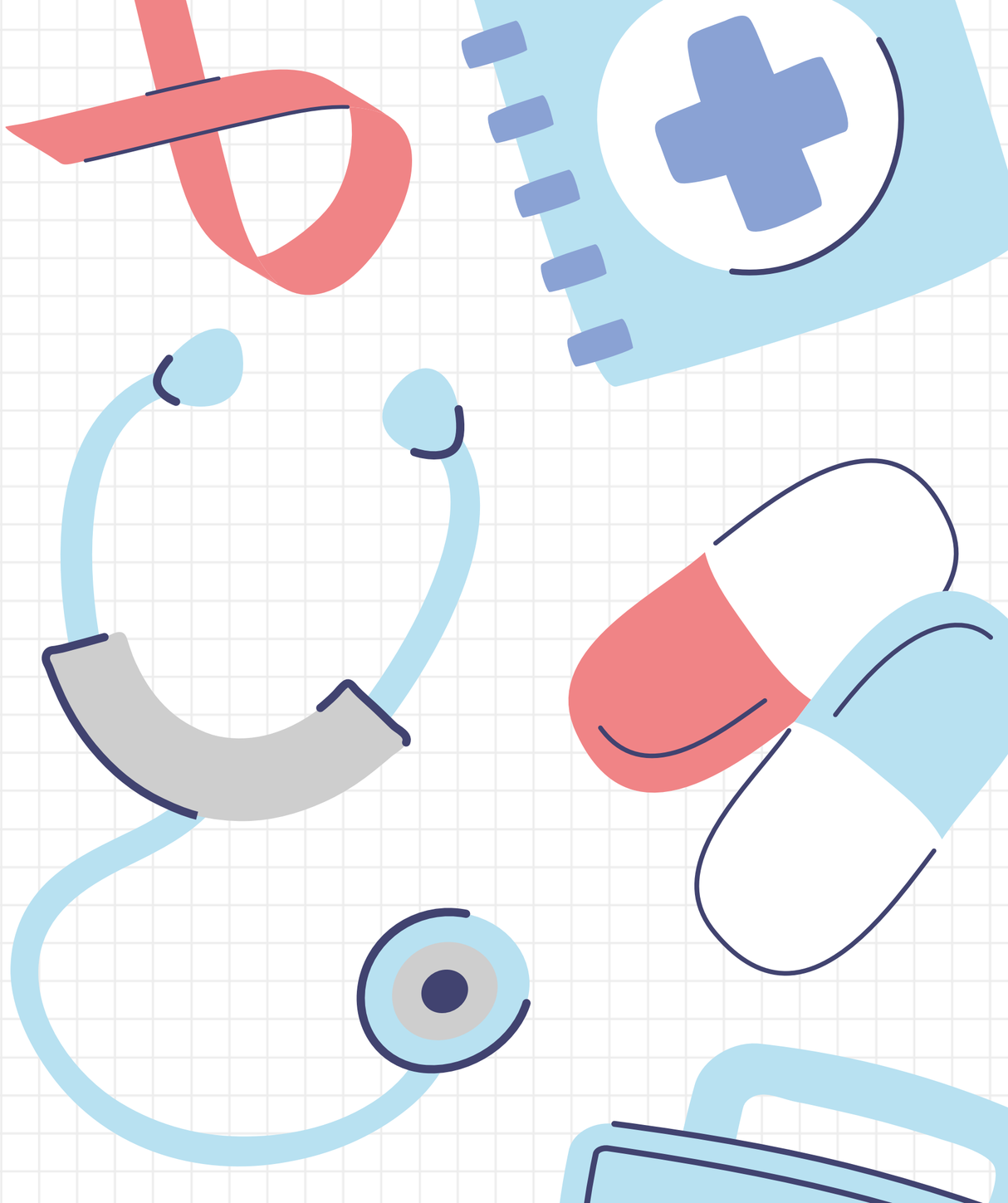
Medsaver addresses a real and widespread problem — millions of patients in India overpay for medicines simply because they lack the knowledge to ask for cheaper alternatives. By combining Vision AI, intelligent drug extraction, and a verified government database, we've built a solution that is frictionless, unbiased, and genuinely impactful.



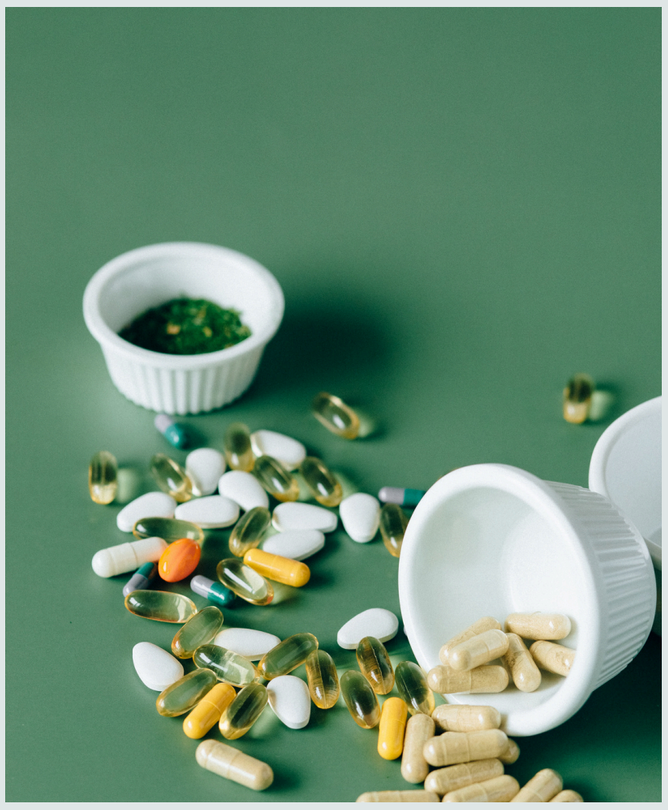
The pipeline is proven, the technology is scalable, and the problem is massive. Medsaver has the potential to democratise access to affordable healthcare for every patient in India — not just the ones who know the right questions to ask.

Thank you!

Team DeepMinds



MEDSAVER



OVERVIEW

- Patient uploads a prescription photo.
- Vision LLM reads the prescription image.
- LLM extracts all medicines with their salt compositions.
- Searches Jan Aushadhi DB and A_Z dataset for matching generics, cheaper alternatives
- Patient sees generic name, pack size, MRP and per-unit price.

OBJECTIVE

Branded medicines in India are 5-10x **more expensive than their generic equivalents**. Medsaver helps patients find the same salt composition at a fraction of the cost by scanning their prescription using AI.

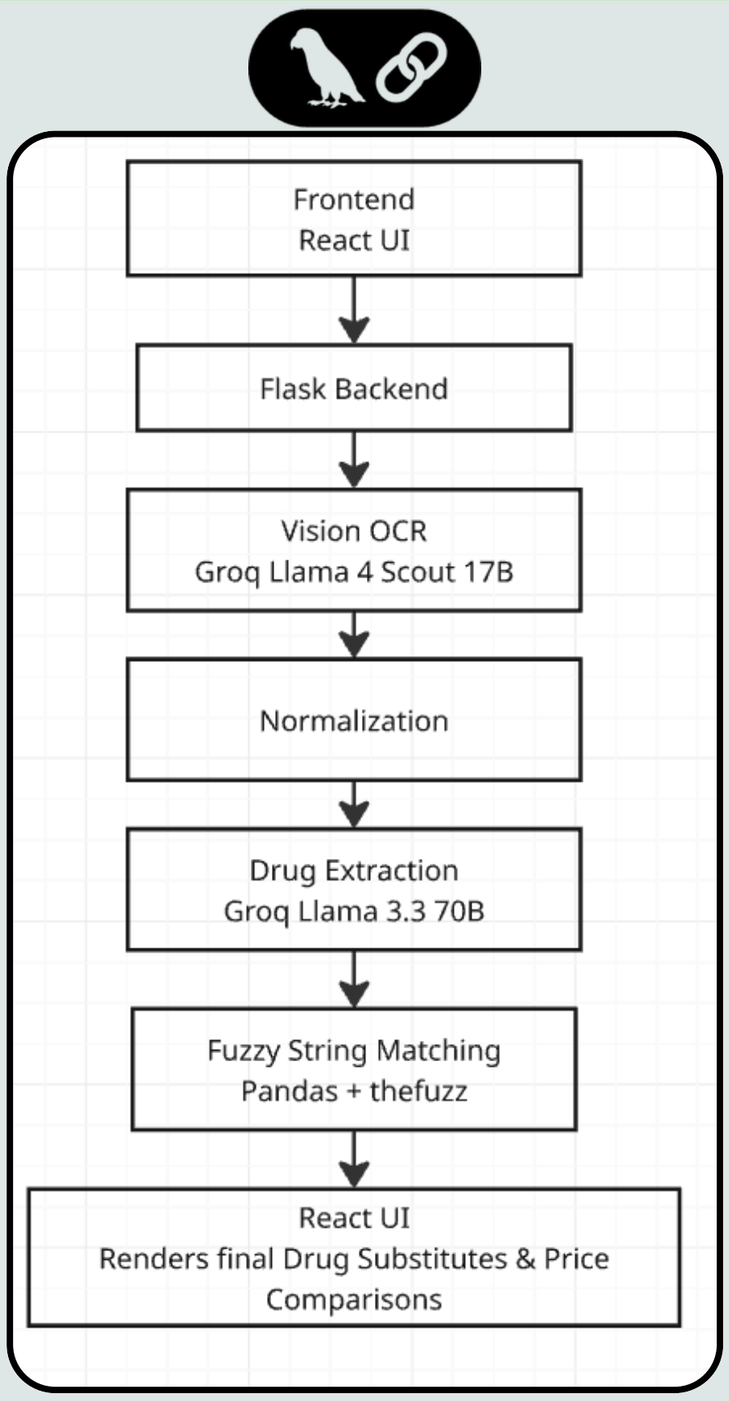
METHODOLOGY

This system converts a prescription image → structured drug data → affordable generic alternatives using AI.

- AI-Powered OCR
- Uses **Vision LLM (Llama 4 Scout)** to read handwritten and printed prescriptions without traditional OCR.
- Smart Drug Extraction
- **Llama 3.3** extracts structured details like brand name, salt composition, dosage, and drug class using typed outputs.
- Intelligent **Generic Matching**
- Matches extracted drugs with **Jan Aushadhi database** and A- Z dataset using:
  - **Salt normalization**
  - **Fuzzy matching** (handles spelling variations)
  - Strength-aware filtering
  - **Scored ranking** (Exact / Good / Partial)
- End-to-End Pipeline
  - Flask APIs + React UI provide a seamless flow from upload → analysis → results.

RESULTS

- **Cost Reduction** as it identifies cheaper generic alternatives → helps users save significantly on medicines.
- Handles **Real-World Prescriptions:** Works with messy handwriting, mixed languages, and unclear dosage.
- **Automation of Manual Work** that is eliminates need for pharmacists / patients to manually search drug equivalents.
- **India-Focused Solution** : Tailored for Indian brands and Jan Aushadhi generics ecosystem.
- **High Matching Accuracy** : Combines LLM intelligence + fuzzy logic → robust matching even with imperfect input.



CONCLUSION

- Enables **affordable medicine** access via generic alternatives
- Converts unstructured prescriptions → structured data
- Demonstrates real-world **impact of LLMs in healthcare**
- **Scalable** with better datasets and integrations

TEAM DEEPMINDS

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